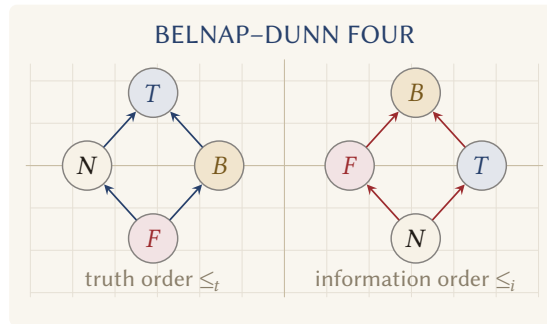




# CourseNote: The “Humbug” of Too Many Values (2026SS)

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## ■ 1 PRELIMINARIES

**1.1 What is Logic?** According to Graham Priest (2014; 2023), logic is about what follows from what, and why. But logic cannot tell us why something is not the case, it only cares about inference. From this approach, logic is about validity: it constructs an artificial metalinguistic system and studies what we can infer from it. However, an opposing argument is that if logic is just about what follows from what and why, then logic is identical to metalogic: a collection of metalinguistic sentences that describe logical consequence  $\Gamma \models A$  (Timothy Williamson, 2022). But metalogic is only part of logic. According to Tim Williamson, logic concerns abstract metaphysics that expresses “a very general structural law about the mostly non-linguistic world.” In that sense, logic is not a neutral linguistic framework but is about the world.

Object-language vs. meta-language: Tarski (1956) makes a distinction between object-language and meta-language. The object-language is the language we are talking about, while the meta-language is the language we use to describe the object-language.

A basic logical system: formal language + proof system + (semantics)

To define formal language, a recursive definition will be introduced.

Bad example: MIU calculus (Hofstadter, Gödel, Escher, Bach), missing semantical part (model theory)?

MIU: formal language + axioms + rules.

- **Base:** if  $X \in \{I, U\}$ , then  $MX$  is a well-formed formula (wff) of MIU.
- **Steps:** if  $MX$  is a wff of MIU, then  $MXI$  is a wff and  $MXU$  is a wff.
- **Axiom:**  $MI$ .

**1.2 What is Classical Logic?** Two designed truth-value: true and false

Law of Non-Contradiction (*principium non-contradictionis*):  $\neg(A \wedge \neg A)$ .

Law of Excluded Middle (*tertium non datur*):  $A \vee \neg A$ .

\*Principle of Explosion (*ex falso quodlibet*):  $(A \wedge \neg A) \rightarrow B$ .

Language:  $L_{cl} = \{\neg, \wedge, \vee, \rightarrow, \perp\}$

Propositional variables:  $p, q, r, p_1, p_2, \dots$

Steps:

- If  $A$  is a formula, then  $\neg A$  is a formula.
- If  $A$  and  $B$  are formulas, then  $(A \wedge B)$  is a formula.
- If  $A$  and  $B$  are formulas, then  $(A \vee B)$  is a formula.
- If  $A$  and  $B$  are formulas, then  $(A \rightarrow B)$  is a formula.

BNF form:

$$A ::= p \mid \neg A \mid (A \wedge A) \mid (A \vee A) \mid (A \rightarrow A)$$

Truth-table:

|        |   |          |   |   |        |   |   |               |   |   |
|--------|---|----------|---|---|--------|---|---|---------------|---|---|
| $\neg$ |   | $\wedge$ | 1 | 0 | $\vee$ | 1 | 0 | $\rightarrow$ | 1 | 0 |
| 1      | 0 | 1        | 1 | 0 | 1      | 1 | 1 | 1             | 1 | 0 |
| 0      | 1 | 0        | 0 | 0 | 0      | 1 | 0 | 0             | 1 | 1 |

Value function and semantic consequence relation:

A Cl-valuation  $v$  is a function that assigns either 0 or 1 to every formula of  $L_{Cl}$  and satisfies the following clauses:

$$\begin{aligned}
 v(\top) &= 1, \\
 v(\perp) &= 0, \\
 v(A \wedge B) &= 1 \iff v(A) = 1 \text{ and } v(B) = 1, \\
 v(A \vee B) &= 1 \iff v(A) = 1 \text{ or } v(B) = 1, \\
 v(\neg A) &= 1 \iff v(A) = 0, \\
 v(A \rightarrow B) &= 1 \iff v(A) = 0 \text{ or } v(B) = 1.
 \end{aligned}$$

A formula  $A$  is **valid** in Cl iff  $v(A) = 1$  for every Cl-valuation  $v$ .

Semantic consequence relation:

$$\Gamma \models_{Cl} A \quad \text{iff} \quad \text{for every Cl-valuation } v, \text{ if } v(B) = 1 \text{ for all } B \in \Gamma, \text{ then } v(A) = 1.$$

(Syntactic Consequence Relation)

### 1.3 What is a Truth-Value?

- Platonism: truth-value as abstract objects (cf. Frege; Church)
- Assign numbers for convenience (designed values): truth-values as algebraic values introduced by a semantic model to represent how sentences behave in inference. In matrix semantics, a formula  $A$  counts as true just in case its value belongs to the set of designated values, i.e.  $v(A) \in D$
- Truth Degree: truth values are degrees of truth. In fuzzy logic, statements take values in  $[0,1]$ , the continuous real interval from 0 to 1, 0 means completely false, 1 means completely true, and intermediate values represent partial truth. This is used to model vagueness. (cf. Gottwald; Zadeh; Goguen)
- Values that are preserved in valid inferences

How Many Values Do We Need? We do not know, may be two or infinite.

Suszko's Thesis (1977): every structural Tarskian many-valued propositional logic can be provided with a bivalent semantics

### 1.4 Many-Valued Logic: Examples, Applications, and Motivation

Examples: LP, K3, FDE,

and fuzzy logic.

Three-valued truth tables with values 1,  $\frac{1}{2}$ , and 0:

|               |               |               |   |               |   |               |               |               |               |
|---------------|---------------|---------------|---|---------------|---|---------------|---------------|---------------|---------------|
| $\wedge$      | 1             | $\frac{1}{2}$ | 0 | $\vee$        | 1 | $\frac{1}{2}$ | 0             | $\neg$        |               |
| 1             | 1             | $\frac{1}{2}$ | 0 | 1             | 1 | 1             | 1             | 1             | 0             |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 0             | 0             | 0             | 0 | 0             | 1 | $\frac{1}{2}$ | 0             | 0             | 1             |

Law of Excluded Middle in the three-valued setting:

| $A$           | $\neg A$      | $A \vee \neg A$ |
|---------------|---------------|-----------------|
| 1             | 0             | 1               |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$   |
| 0             | 1             | 1               |

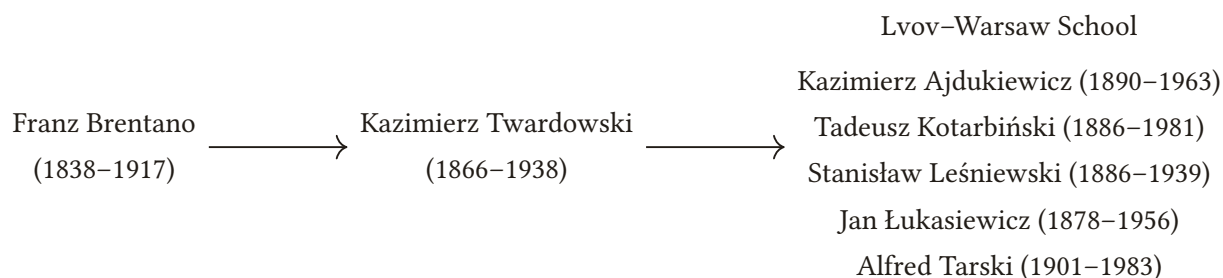
Validity: Under  $D = \{1, \frac{1}{2}\}$ ,  $A \vee \neg A$  is valid, since its value is always designated. Under  $D = \{1\}$ ,  $A \vee \neg A$  is not valid, since for  $A = \frac{1}{2}$  its value is only  $\frac{1}{2}$ .

Application: Washing machines are a real-world application of fuzzy logic.

Why it matters:

- future contingent propositions, such as “Tomorrow Daniel will eat pasta,”
- self-referential paradoxes, such as “This sentence is false.”
- vagueness, sort of paradox

## 1.5 History: Jan Łukasiewicz (1878-1956)



Jan Łukasiewicz was a Polish logician and philosopher born in Lwów in 1878. He studied philosophy at the University of Lwów and later became one of the leading figures of the Lwów-Warsaw School. He is especially known for his work on many-valued logic, Polish notation, modal logic, and the history of logic. In 1915, he became a professor at the University of Warsaw. He also served twice as rector of the University of Warsaw. After the outbreak of the Second World War, his academic life in Poland was disrupted. After the war, he settled in Dublin, where he worked at the Royal Irish Academy and continued his research in logic. He died in Dublin in 1956.

In 1918, in his farewell lecture, Łukasiewicz talked about his main motivation for developing three-valued logic, which was to solve the problem of determinism. The ancient idea is that “all events and phenomena are interconnected by causal links and follow from one another as theorems in a scientific theory” (Łukasiewicz, 1970, p. 84). But Łukasiewicz did not want to be restricted by that, and he used the passive word “coercion” to indicate that human beings are forced to follow physical and logical laws. The point is: where is the place of the human creative mind and the possibility of humans affecting the course of events? He proposed two solutions: one is skepticism and abandoning research, and the other is to reexamine the idea that science is based on Aristotelian logic. Łukasiewicz chose the second path. First, the principle of non-contradiction is not self-evident. Second, there are possible propositions that can be assigned a third truth-value. This is related to what he called “objective possibility,” and a good example is a future contingent proposition like “I will go to Amsterdam tomorrow.” This proposition is indeterminate and can be both true and false.

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## 2 MATRICES AND ABSTRACT LOGICS

**Definition 1.** A logic is a pair  $\mathcal{L} = \langle S, \vdash \rangle$ , where  $S$  is a non-empty set and  $\vdash \subseteq \mathcal{P}(S) \times S$ . We call  $\vdash$  a **Tarskian consequence relation** if it satisfies the following conditions for all  $\Gamma, \Delta \subseteq S$  and all  $A \in S$ :

- **Reflexivity:** if  $A \in \Gamma$ , then  $\Gamma \vdash A$ .
- **Monotonicity:** if  $\Gamma \vdash A$  and  $\Gamma \subseteq \Delta$ , then  $\Delta \vdash A$ .
- **Cut:** if  $\Gamma \vdash B$  for every  $B \in \Delta$ , and  $\Delta \vdash A$ , then  $\Gamma \vdash A$ .

If  $\vdash$  satisfies these conditions, then  $\mathcal{L}$  is called a **Tarskian logic**.

**Definition 2.** An **algebra** is defined as  $\mathcal{A} = \langle A, o_1, \dots, o_n \rangle$ , where  $A$  is a non-empty set and  $o_1, \dots, o_n$  are operations defined on  $A$  that are **closed** under  $A$ : if  $o_i$  is a  $k$ -ary operation on  $A$  (a function from  $A^k$  to  $A$  and  $x_1, \dots, x_k \in A$ , then  $o_i(x_1, \dots, x_k) \in A$ ). The **arity** of a connective or operation is the number of elements it takes as inputs. If a logical connective or operation  $x_i$  operates on  $k$  elements, then its arity is  $k$ , usually written as  $\delta(x_i) = k$ . The **signature** of an algebra records the arities of the operations occurring in it. If  $\delta(o_i)$  denotes the arity of the operation  $o_i$ , then the signature of the algebra  $\mathcal{A}$  is defined as the  $n$ -tuple  $\langle \delta(o_1), \dots, \delta(o_n) \rangle$ .

*Example 1. Sentential Algebra (Satzalgebra):* The propositional language can be viewed algebraically as an algebra of formulas, for example  $\mathcal{S} = \langle \text{For}, \neg, \wedge, \vee, \rightarrow \rangle$ , whose signature is  $\langle 1, 2, 2, 2 \rangle$ . More generally, we can define  $\mathcal{S} = \langle \text{For}, c_1^{k_1}, c_2^{k_2}, \dots, c_n^{k_n} \rangle$ , where For satisfies the following recursive clauses: the set  $V$  of sentential variables, or atomic formulas, is a subset of For, i.e.  $V \subseteq \text{For}$ ; if  $k_i = 0$ , then  $c_i^{k_i}$  is a constant and  $c_i^{k_i} \in \text{For}$ ; if  $k_i \geq 1$ ,  $c_i^{k_i}$  is a  $k_i$ -ary connective, and  $A_1, \dots, A_{k_i} \in \text{For}$ , then  $c_i^{k_i}(A_1, \dots, A_{k_i}) \in \text{For}$ ; and nothing else belongs to For.

*Example 2.*  $\mathcal{CL}_1 = \langle \{1, 0\}, 1 - x, \min(x, y), \max(x, y) \rangle$

*Example 3.*  $\mathcal{CL}_2 = \langle \{1, 0\}, 1 - x, x \cdot y, x + y - x \cdot y \rangle$

*Example 4.*  $\mathcal{K}_3 = \langle \{1, \frac{1}{2}, 0\}, 1 - x, \min(x, y), \max(x, y) \rangle$

*Example 5.*  $\mathcal{A} = \langle \{1, \frac{1}{2}, 0\}, 1 - x, x \cdot y, x + y - x \cdot y \rangle$  is not an algebra. Let  $x = \frac{1}{2}$  and  $y = \frac{1}{2}$ , then  $x + y - x \cdot y = \frac{3}{4} \notin \{1, \frac{1}{2}, 0\}$ .

**Definition 3.** Two algebras  $\mathcal{A} = \langle A, o_1, \dots, o_n \rangle$  and  $\mathcal{B} = \langle B, p_1, \dots, p_n \rangle$  are called **similar** iff their signatures are identical, i.e.

$$\langle \delta(o_1), \dots, \delta(o_n) \rangle = \langle \delta(p_1), \dots, \delta(p_n) \rangle.$$

In other words, corresponding operations in the two algebras have the same arities.

**Definition 4.** Let  $\mathcal{A} = \langle A, o_1, \dots, o_n \rangle$  and  $\mathcal{B} = \langle B, p_1, \dots, p_n \rangle$  be similar algebras. A **morphism** from  $\mathcal{A}$  to  $\mathcal{B}$  is a map  $h : A \rightarrow B$ . A morphism  $h$  is called a **homomorphism** iff it preserves the operations of the algebra, i.e. for every  $k$ -ary operation and all  $x_1, \dots, x_k \in A$ ,

$$h(o_j(x_1, \dots, x_k)) = p_j(h(x_1), \dots, h(x_k)).$$

In other words, the value of a complex expression is preserved under the map  $h$ .

*Example.* An **evaluation (valuation, or assignment)** is a homomorphism from the sentential algebra  $\mathcal{S}$  to a logical algebra  $\mathcal{L}$ . For example, let  $\mathcal{S} = \langle \text{For}, \neg, \wedge, \vee \rangle$  and  $\mathcal{L}' = \langle \{0, 1\}, 1 - x, \min(x, y), \max(x, y) \rangle$ . Then a valuation is a homomorphism  $v : \text{For} \rightarrow \{0, 1\}$ , like  $v(A \wedge B) = \min(v(A), v(B))$ .

**Lemma.** If  $h : \mathbf{A} \rightarrow \mathbf{B}$  and  $g : \mathbf{B} \rightarrow \mathbf{C}$  are homomorphisms between similar algebras, then  $g \circ h : \mathbf{A} \rightarrow \mathbf{C}$  is a homomorphism.

*Proof.* Write  $\mathbf{A} = \langle A, o_1, \dots, o_n \rangle$ ,  $\mathbf{B} = \langle B, p_1, \dots, p_n \rangle$ , and  $\mathbf{C} = \langle C, q_1, \dots, q_n \rangle$ . If  $o_i$  has arity  $m$ , then for all  $x_1, \dots, x_m \in A$ ,

$$\begin{aligned} (g \circ h)(o_i(x_1, \dots, x_m)) &= g(p_i(h(x_1), \dots, h(x_m))) \\ &= q_i((g \circ h)(x_1), \dots, (g \circ h)(x_m)). \end{aligned}$$

Thus  $g \circ h$  preserves every operation. ■

**Definition 5.** Let  $\mathcal{A}$  and  $\mathcal{B}$  be similar algebras. An **isomorphism** from  $\mathcal{A}$  to  $\mathcal{B}$  is a homomorphism  $h : A \rightarrow B$  that is bijective. That is:

- **Surjectivity:** for every  $b \in B$ , there is an  $a \in A$  such that  $h(a) = b$ .
- **Injectivity:** if  $a, a' \in A$  and  $a \neq a'$ , then  $h(a) \neq h(a')$ .

**Definition 6.** An **endomorphism** of an algebra  $\mathcal{A}$  is a homomorphism from  $\mathcal{A}$  to itself, i.e. a map  $h : A \rightarrow A$  that preserves all operations of the algebra. So, for every  $k$ -ary operation  $o_j$  and all  $x_1, \dots, x_k \in A$ ,

$$h(o_j(x_1, \dots, x_k)) = o_j(h(x_1), \dots, h(x_k)).$$

**Definition 7.** Let  $\mathcal{S} = \langle \text{For}, c_1^{k_1}, c_2^{k_2}, \dots, c_n^{k_n} \rangle$  be a sentential algebra. A **substitution** is an endomorphism of  $\mathcal{S}$ , i.e. a map  $e : \text{For} \rightarrow \text{For}$  such that for every connective  $c_i^{k_i}$  and all formulas  $A_1, \dots, A_{k_i} \in \text{For}$ ,  $e(c_i^{k_i}(A_1, \dots, A_{k_i})) = c_i^{k_i}(e(A_1), \dots, e(A_{k_i}))$ .

*Example.* Consider the sentential algebra  $\mathcal{S} = \langle \text{For}, \neg, \wedge, \vee, \rightarrow \rangle$ . Define a substitution  $e : \text{For} \rightarrow \text{For}$  by assigning  $e(p) = p \wedge q$  and  $e(q) = r$  to sentential variables, and then extending this assignment homomorphically to all formulas. Then, for instance,  $e(p \rightarrow \neg q) = (p \wedge q) \rightarrow \neg r$ .

**Definition 8.** Let  $\mathcal{L} = \langle S, \vdash \rangle$  be a logic. We say that  $\vdash$  is a **structural finite Tarskian consequence relation** iff:

- $\vdash$  is a Tarskian consequence relation, i.e. it satisfies reflexivity, monotonicity, and cut;
- $\vdash$  is **finite** (or finitary): if  $\Gamma \vdash A$ , then there is a finite set  $\Delta \subseteq \Gamma$  such that  $\Delta \vdash A$ ;
- $\vdash$  is **structural**: if  $\Gamma \vdash A$ , then  $e[\Gamma] \vdash e(A)$  for every substitution  $e \in \text{End}(S)$ , where  $e[\Gamma] = \{e(B) \mid B \in \Gamma\}$ .

**Definition 9.** A **matrix** is a structure  $M = \langle \mathcal{A}, D \rangle$ , where  $\mathcal{A} = \langle A, o_1, \dots, o_n \rangle$  is an algebra and  $D$  is a non-empty subset of  $A$ . The set  $D$  is called the set of **designated values**.

**Definition 10.** Let  $\mathcal{S} = \langle \text{For}, c_1^{k_1}, c_2^{k_2}, \dots, c_n^{k_n} \rangle$  be a sentential algebra and let  $M = \langle \mathcal{A}, D \rangle$  be a matrix such that  $\mathcal{A}$  is similar to  $\mathcal{S}$ . The **matrix semantic consequence relation**  $\models_M \subseteq \mathcal{P}(\text{For}) \times \text{For}$  is defined as follows:

$\Gamma \models_M A$  iff for every homomorphism  $h \in \text{Hom}(\mathcal{S}, \mathcal{A})$ , if  $h(B) \in D$  for all  $B \in \Gamma$ , then  $h(A) \in D$

In words:  $A$  is a semantic consequence of  $\Gamma$  in the matrix  $M$  iff every valuation that assigns designated values to all formulas in  $\Gamma$  also assigns a designated value to  $A$ . Additionally, every logic satisfying a Tarskian consequence relation can be defined by means of matrix semantics.

**Fact.** The matrix consequence relation is a structural Tarskian consequence relation.

*Proof.* **Reflexivity.** Suppose  $A \in \Sigma$  but  $\Sigma \not\models_M A$ . Then there is a valuation  $v : \text{For} \rightarrow \mathbf{A}$  such that  $v(C) \in D$  for every  $C \in \Sigma$ , while  $v(A) \notin D$ . Since  $A \in \Sigma$ , this is a contradiction. **Monotonicity.** Suppose  $\Sigma \models_M A$  but  $\Sigma \cup \{B\} \not\models_M A$ . Then there is a valuation  $v$  such that  $v(C) \in D$  for every  $C \in \Sigma \cup \{B\}$ , while  $v(A) \notin D$ . In particular,  $v(C) \in D$  for every  $C \in \Sigma$ , contradicting  $\Sigma \models_M A$ . **Cut.** Suppose  $\Sigma \models_M A$  and  $\Delta \cup \{A\} \models_M B$ , but  $\Sigma \cup \Delta \not\models_M B$ . Then there is a valuation  $v$  such that  $v(C) \in D$  for every  $C \in \Sigma \cup \Delta$ , while  $v(B) \notin D$ . Since  $v$  designates every member of  $\Sigma$  and  $\Sigma \models_M A$ , we have  $v(A) \in D$ . Hence  $v$  designates every member of  $\Delta \cup \{A\}$ , so  $\Delta \cup \{A\} \models_M B$  gives  $v(B) \in D$ , a contradiction. **Structurality.** Suppose  $\Sigma \models_M A$ , and let  $v$  be any valuation that designates every member of  $e[\Sigma]$ . Since  $e$  is an endomorphism and  $v$  is a homomorphism,  $v \circ e$  is a

homomorphism by the preceding exercise. For every  $C \in \Sigma$ , we have  $e(C) \in e[\Sigma]$ , and therefore  $(v \circ e)(C) = v(e(C)) \in D$ . Thus  $v \circ e$  designates every member of  $\Sigma$ . From  $\Sigma \models_M A$  it follows that  $(v \circ e)(A) = v(e(A)) \in D$ . Therefore  $e[\Sigma] \models_M e(A)$ . ■

**Semantics** What is a semantics? It usually consists of three things:

1. a class of models or structures,  $\mathcal{M}$ ;
2. evaluations, valuations, or assignments,  $E_m$ ;
3. a satisfaction relation,  $\models$  or  $\vDash$ , for example  $m \models \varphi[v]$ , where  $m \in \mathcal{M}$ ,  $\varphi \in \text{For}$ , and  $v \in E_m$ .

Truth-preserving logics vs. logics preserving degrees of truth:

Truth-preserving: whenever all premises are true, the conclusion is true as well. This is what we use to define the usual semantic consequence relation:  $A$  follows from  $\Gamma$  iff, for every evaluation, if every  $B \in \Gamma$  is true (designated), then  $A$  is also true (designated).

Truth-degree preserving:  $A$  follows from  $\Gamma$  iff, for every evaluation, every truth degree attained by all formulas  $B \in \Gamma$  is also attained by  $A$ .

The relation between syntax and semantics: according to Montague (1970), syntax is an algebra, semantics is an algebra, and there is a homomorphism mapping elements of the syntactic algebra to elements of the semantic algebra. This is also called the principle of compositionality.

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## ■ 3 FUZZY LOGIC

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**3.1 Problem of Vagueness** There are vague predicates such as “bald,” “tall,” and “old.” It is difficult to provide precise standards for applying them, and they have borderline cases. Philosophers are interested in vagueness mainly because of the sorites paradox, while scientists may be interested in vagueness as a natural phenomenon to be investigated in complex systems.

*Sorites Paradox:* the sorites paradox was first introduced in the history of philosophy by the ancient Greek philosopher Eubulides of Miletus (around the 4th century BC). He proposed seven puzzles, two of which are relevant to the sorites paradox: the heap puzzle and the bald man puzzle.

**Heap Puzzle** How many grains of wheat make a heap?

1. One grain does not make a heap.
2. Two grains do not make a heap.
3. ...
4. If  $n$  grains of wheat do not make a heap, then  $n + 1$  grains of wheat also do not make a heap.

**Bald Man Puzzle.** How many hairs can a bald man have?

1. A man with one hair is bald.
2. A man with two hairs is bald.
3. ...
4. If a man with  $n$  hairs is bald, then a man with  $n + 1$  hairs is also bald.

Besides these discrete and numerical sorites cases, there are also continuous cases, such as the phenomenal sorites paradox.

**Red Perception Puzzle.** At what point does a colour patch cease to look red? Consider a continuous colour spectrum from red to yellow. Let  $P_x$  be the colour patch at position  $x$  on the spectrum, where  $x \in [0, 1]$ .  $P_0$  looks clearly red, and  $P_1$  looks clearly yellow.

1.  $P_0$  looks red.
2. If  $P_x$  looks red, and  $P_x$  looks the same in respect of colour as  $P_y$ , then  $P_y$  also looks red.
3. For any sufficiently small difference between  $x$  and  $y$ ,  $P_x$  looks the same in respect of colour as  $P_y$ .
4. Therefore, for any  $x \in [0, 1]$ ,  $P_x$  looks red.
5. Therefore,  $P_1$  looks red.

**3.2 Fuzzy Set** Lotfi A. Zadeh (1921–2017) is usually considered the founder of fuzzy logic. In 1965, he introduced the concept of a fuzzy set. One motivation for Zadeh was that classical mathematics seemed inadequate for dealing with highly complex systems. He observed that systems such as biological systems are difficult to describe in precise mathematical language. This led him to question classical set theory, according to which an object either belongs to a set or does not belong to it. Classical set theory assumes sharp boundaries between members and non-members, but this is not the case for many real-world phenomena. Zadeh's idea was to replace the strict distinction between membership and non-membership with degrees of membership. A fuzzy set is a class whose members can belong to it to different degrees.

**Definition 1.** The **characteristic function** (or **indicator function**) of a set  $A$  is defined by

$$\chi_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

If  $A \subseteq B$ , then  $\chi_A(x) \leq \chi_B(x)$  for every  $x \in U$ .

**Operations on characteristic functions.**

$$\begin{aligned} \chi_{A \cup B}(x) &= \chi_A(x) \vee \chi_B(x) = \max(\chi_A(x), \chi_B(x)), \\ \chi_{A \cap B}(x) &= \chi_A(x) \wedge \chi_B(x) = \min(\chi_A(x), \chi_B(x)), \\ \chi_{\bar{A}}(x) &= 1 - \chi_A(x). \end{aligned}$$

A crisp set permits only the two membership values 0 and 1. A fuzzy set replaces this sharp alternative with the whole interval  $[0, 1]$ .

**Definition 2.** A **fuzzy set** on a set  $U$  is a function  $\mu : U \rightarrow [0, 1]$ .

**Definition 3.** When the fuzzy set is denoted by  $A$ , its **membership function** is written

$$\mu_A : U \longrightarrow [0, 1].$$

The number  $\mu_A(x)$ , also written  $A(x)$ , is the **degree of membership** of  $x$  in  $A$ . Thus  $\mu_A(x) \in [0, 1]$ .

Two fuzzy sets  $A$  and  $B$  are equal iff  $\mu_A(x) = \mu_B(x)$  for every  $x \in U$ .

**Operations on Fuzzy Logic** For fuzzy sets  $A$  and  $B$  on  $U$ , the standard intersection, union, and complement are defined pointwise by

$$\begin{aligned} \mu_{A \cap B}(x) &= \min(\mu_A(x), \mu_B(x)), \\ \mu_{A \cup B}(x) &= \max(\mu_A(x), \mu_B(x)), \\ \mu_{\bar{A}}(x) &= 1 - \mu_A(x). \end{aligned}$$

### 3.3 t-norms

**Definition. 4** A binary operation

$$T : [0, 1] \times [0, 1] \longrightarrow [0, 1]$$

is a **t-norm** iff, for all  $x, y, z \in [0, 1]$ , it satisfies:

1.  $T(x, 1) = x$ . (1 is its unit element)
2.  $T(x, y) = T(y, x)$ . (Commutativity)
3.  $T(T(x, y), z) = T(x, T(y, z))$ . (Associativity)
4. If  $x \leq y$ , then  $T(x, z) \leq T(y, z)$ . (Monotonicity)

*Example 1.* The minimum t-norm, also called the **Gödel t-norm**, is

$$T_1(x, y) = \min(x, y).$$

*Example 2.* The **Łukasiewicz t-norm** is

$$T_2(x, y) = \max(0, x + y - 1).$$

*Example 3.* The **product t-norm** is

$$T_3(x, y) = xy.$$

*Example 4.*

$$T_4(x, y) = x \wedge y.$$

*Example 5.*

$$T_5(x, y) = \frac{xy}{x + y - xy},$$

where the value at  $(0, 0)$  is defined to be 0.

*Example 6.* The **drastic t-norm** is

$$T_6(x, y) = \begin{cases} x \wedge y, & \text{if } x \vee y = 1, \\ 0, & \text{otherwise.} \end{cases}$$

## ■ 4 Ł3, K3 AND LP

**4.1 Ł3** Łukasiewicz added a third value, “1/2,” to represent “possible” and hoped to discuss the truth value of future contingent sentences. For example, a proposition  $p$  is possible iff it is not false. The truth value of  $p$  could be 1 or 1/2.

**Definition 1.** The three-valued Łukasiewicz logic **Ł3** is based on

$$\mathbf{L}_3 = \langle \{1, 1/2, 0\}, \{\neg, \wedge, \vee, \rightarrow_{\mathbf{L}_3}\}, \{1\} \rangle,$$

Negation and implication are defined as:

$$\neg x = 1 - x \quad x \rightarrow_{\mathbf{L}_3} y = \min(1, 1 - x + y).$$

Conjunction and Disjunction can be defined as:

$$x \wedge y = \min(x, y) \quad x \vee y = \max(x, y)$$

| $x$           | $\neg x$      | $\rightarrow_{\mathbf{L}_3}$ | 1 | 1/2           | 0             |
|---------------|---------------|------------------------------|---|---------------|---------------|
| 1             | 0             | 1                            | 1 | $\frac{1}{2}$ | 0             |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$                | 1 | 1             | $\frac{1}{2}$ |
| 0             | 1             | 0                            | 1 | 1             | 1             |

We can notice that the implication in Ł3 is not a material conditional. One reason for this is to preserve  $(p \rightarrow p)$  as a tautology in the logic.

**Proposition 1.**  $\models_{\mathbf{L}_3} p \rightarrow p$ ,

*Proof.* We can simply write the truth table for  $p \rightarrow p$ . ■

**Proposition 2.**  $A, A \rightarrow B \models_{\mathbf{L}_3} B$ .

*Proof.* Suppose the evaluations of  $A$  and  $A \rightarrow B$  are designated, that is, both are 1. Then we can easily see that the evaluation of  $B$  is also designated, that is, it is 1. ■

**Proposition 3.**  $\Gamma \models_{\mathbf{L}_3} A \implies \Gamma \models_{\mathbf{CPL}} A$ .

*Proof.* Suppose that  $\Gamma \models_{\mathbf{L}_3} A$ . Let  $v_{\mathbf{CPL}}$  be an arbitrary classical valuation such that  $v_{\mathbf{CPL}}(B) = 1$  for every  $B \in \Gamma$ . Since  $\{0, 1\} \subseteq \{0, \frac{1}{2}, 1\}$ , we can construct  $v_{\mathbf{L}_3}$  such that  $v_{\mathbf{CPL}}(p) = v_{\mathbf{L}_3}(p)$ . Easily, by induction, we actually have  $v_{\mathbf{CPL}}(H) = v_{\mathbf{L}_3}(H)$  for every  $H \in \text{For}$ . Therefore, since  $v_{\mathbf{CPL}}(B) = 1$ , we have  $v_{\mathbf{L}_3}(B) = 1$ . Because  $\Gamma \models_{\mathbf{L}_3} A$ ,  $v_{\mathbf{L}_3}(A) = 1$ . Then we have  $v_{\mathbf{CPL}}(A) = 1$ . Hence,  $\Gamma \models_{\mathbf{CPL}} A$ . ■

We can see that modus ponens is a valid inference in Ł3, and many classical tautologies are also tautologies in Ł3. However, some classical tautologies are not valid in Ł3, such as the law of excluded middle  $(p \vee \neg p)$  and the law of non-contradiction  $(\neg(p \wedge \neg p))$ . One interesting result is

that  $\mathbb{L}_3$  is a sublogic of classical propositional logic: it does not generate any classically invalid inferences.

**Proposition 4.**  $p \wedge \neg p \vDash_{\mathbb{L}_3} q$ .

*Proof.* The valuation of  $p \wedge \neg p$  belongs to  $\{0, \frac{1}{2}\}$ , so  $v(p \wedge \neg p) \notin \{1\}$ . Then, no matter what the valuation of  $q$  is,  $p \wedge \neg p \vDash_{\mathbb{L}_3} q$  is always valid. ■

**4.2 K3** In contrast to Łukasiewicz's philosophical motivation, Stephen Cole Kleene (1909–1994) proposed two kinds of three-valued logic in order to solve problems in computation theory (partial recursive predicates). A computational procedure may have three possible outcomes for a given input:

1. The computation terminates and returns “true.”
2. The computation terminates and returns “false.”
3. The computation never terminates, so no true or false result is obtained.

In the third case, the third value, “ $1/2$ ” can be understood as undefined or undecided by computation. Kleene, in his *Introduction to Metamathematics* (1952), introduced two kinds of interpretations of the connectives. The first is based on sequential computation. For example, suppose we input  $P \wedge Q$ , where  $P$  is undefined and  $Q$  is false. The program first processes  $P$  and then  $Q$ , but there will be no result because the program gets stuck on  $P$ . In such a case, the truth value of  $P \wedge Q$  is undefined ( $1/2$ ). The second is based on parallel computation. The program can process  $P$  and  $Q$  at the same time. If  $P$  is undefined and  $Q$  is false, then the program can return the result “false”. Thus, the truth value of  $P \wedge Q$  is false.

**Definition. 2** Strong Kleene Logic is defined as:

$$\mathbf{K}_3 = \langle \{1, \frac{1}{2}, 0\}, \neg, \wedge, \vee, \{1\} \rangle$$

Similar to  $\mathbb{L}_3$ , negation, conjunction, and disjunction can be defined as follows:

$$\neg x = 1 - x \quad x \wedge y = \min(x, y) \quad x \vee y = \max(x, y).$$

Additionally, implication can be added to  $\mathbf{K}_3$  and defined as the material conditional:

$$x \rightarrow y = \neg x \vee y.$$

| $\wedge$      | 1             | $\frac{1}{2}$ | 0 | $\vee$        | 1 | $\frac{1}{2}$ | 0             | $x$           | $\neg x$      | $\rightarrow_K$ | 1 | $\frac{1}{2}$ | 0             |
|---------------|---------------|---------------|---|---------------|---|---------------|---------------|---------------|---------------|-----------------|---|---------------|---------------|
| 1             | 1             | $\frac{1}{2}$ | 0 | 1             | 1 | 1             | 1             | 1             | 0             | 1               | 1 | $\frac{1}{2}$ | 0             |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$   | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 0             | 0             | 0             | 0 | 0             | 1 | $\frac{1}{2}$ | 0             | 0             | 1             | 0               | 1 | 1             | 1             |

The difference between  $\mathbb{L}_3$  and the Strong Kleene algebra is concentrated in the middle case:

$$\frac{1}{2} \rightarrow_{\mathbb{L}_3} \frac{1}{2} = 1, \quad \frac{1}{2} \rightarrow_K \frac{1}{2} = \frac{1}{2}.$$

**Proposition 5.**  $K_3$  have no tautologies.

*Proof.* Suppose that  $\models_{K_3} A$ , this means that  $v(A) = 1$  for every valuation  $v$ . Let  $v_p$  be a valuation such that  $v_p(p) = \frac{1}{2}$  for every atomic formula  $p$ . We can prove by induction that  $v_p(H) = \frac{1}{2}$  for every formula  $H$ . In particular,  $v_p(A) = \frac{1}{2}$ , which contradicts our assumption that  $v(A) = 1$  for every valuation  $v$ . Therefore,  $K_3$  has no tautologies. ■

**Proposition 6.**  $p \wedge \neg p \models_{K_3} q$ .

*Proof.* A similar proof as in prop.4. ■

**4.3 LP** Graham Priest (1979, 1984) proposed the Logic of Paradox (LP) to address logical paradoxes such as the liar paradox. Priest argues that we should accept that some paradoxical propositions are both true and false, while revising the logical consequence relation so that these local contradictions do not entail arbitrary propositions. In that case, Priest revised the set of designated values of the three-valued logic so that  $D = \{1, \frac{1}{2}\}$ . The truth value  $\frac{1}{2}$  can be understood here as both true and false.

**Definition 3.** The Logic of Paradox (LP) is defined as:

$$\mathbf{LP} = \left\langle \left\{1, \frac{1}{2}, 0\right\}, \neg, \wedge, \vee, \left\{1, \frac{1}{2}\right\} \right\rangle.$$

The truth tables for  $\neg$ ,  $\wedge$ , and  $\vee$  are the same as those of  $K_3$  and  $L_3$ .

The implication is defined as material implication, same as in  $K_3$ :  $A \rightarrow_{\mathbf{LP}} B = \neg A \vee B$ .

**Definition 4.** A logic  $L$  is called *paraconsistent* if the principle of explosion does not hold in  $L$ . Formally, there exist formulas  $A$  and  $B$  such that

$$A, \neg A \not\models_L B.$$

**Proposition 7.** LP is paraconsistent.

*Proof.* Let  $v(A) = v(\neg A) = \frac{1}{2}$ , and let  $v(B) = 0$ . ■

**Proposition 8.**  $\models_{\mathbf{LP}} p \vee \neg p$ .

*Proof.* This can be simply verified by the truth-table of LP. ■

**Proposition 9.**  $\Gamma \models_{\mathbf{LP}} A \Rightarrow \Gamma \models_{\mathbf{CL}} A$ .

*Proof.* Suppose that  $\Gamma \models_{\mathbf{LP}} A$ . Let  $v$  be an arbitrary classical valuation such that  $v(B) = 1$  for every  $B \in \Gamma$ . We can prove by structural induction that  $v$  is also a valuation in LP ( $D_{\mathbf{CL}} \subseteq D_{\mathbf{LP}}$ ,  $1 \subseteq \{1, \frac{1}{2}\}$ ). Because  $\Gamma \models_{\mathbf{LP}} A$  and  $v(B) = 1$  for every  $B \in \Gamma$ , we have  $v(A) = 1 \in \{1, \frac{1}{2}\}$ . Hence,  $\Gamma \models_{\mathbf{CL}} A$ . ■

**4.4 Comparison** With  $\vee$  interpreted as maximum, the three logics behave as follows:

| Logic          | Designated values    | $\models A \vee \neg A$ | $A, \neg A \models B$ | $A, A \rightarrow B \models B$ |
|----------------|----------------------|-------------------------|-----------------------|--------------------------------|
| $\mathbb{L}_3$ | $\{1\}$              | No                      | Yes                   | Yes                            |
| $K_3$          | $\{1\}$              | No                      | Yes                   | Yes                            |
| LP             | $\{1, \frac{1}{2}\}$ | Yes                     | No                    | No                             |

$K_3$  is paracomplete. LP is paraconsistent.

|                                                                                                   | $K_3$ | $\mathbb{L}_3$ | LP |
|---------------------------------------------------------------------------------------------------|-------|----------------|----|
| (1) $q \models p \rightarrow q$                                                                   | ✓     | ✓              | ✓  |
| (2) $\neg p \models p \rightarrow q$                                                              | ✓     | ✓              | ✓  |
| (3) $(p \wedge q) \rightarrow r \models (p \rightarrow r) \vee (q \rightarrow r)$                 | ✓     | ✓              | ✓  |
| (4) $(p \rightarrow q) \wedge (r \rightarrow s) \models (p \rightarrow s) \vee (r \rightarrow q)$ | ✓     | ✓              | ✓  |
| (5) $\neg(p \rightarrow q) \models p$                                                             | ✓     | ✓              | ✓  |
| (6) $p \rightarrow r \models (p \wedge q) \rightarrow r$                                          | ✓     | ✓              | ✓  |
| (7) $p \rightarrow q, q \rightarrow r \models p \rightarrow r$                                    | ✓     | ✓              | ✗  |
| (8) $p \rightarrow q \models \neg q \rightarrow \neg p$                                           | ✓     | ✓              | ✓  |
| (9) $\models p \rightarrow (q \vee \neg q)$                                                       | ✗     | ✗              | ✓  |
| (10) $\models (p \wedge \neg p) \rightarrow q$                                                    | ✗     | ✗              | ✓  |

**Exercise.** Let **For** be the formula algebra with carrier **For** and connectives  $\neg, \wedge, \vee$ , and  $\rightarrow$ . Consider  $\mathbf{A} = \langle [0, 1], \neg, \wedge, \vee, \rightarrow \rangle$ ,  $\mathcal{M} = \langle \mathbf{A}, \{1\} \rangle$ , where

$$\neg x = 1 - x, \quad x \wedge y = xy, \quad x \vee y = x + y - xy, \quad x \rightarrow y = \min(1, y/x)$$

when  $x > 0$ , while  $0 \rightarrow y = 1$ . The last clause makes implication defined on the whole unit interval. Does  $\Sigma \models_{\mathcal{M}} A \implies \Sigma \models_{\text{CPL}} A$  hold for every  $\Sigma \cup \{A\} \subseteq \text{For}$ ?

*Proof.* We prove the contrapositive. Suppose that  $\Sigma \not\models_{\text{CPL}} A$ . There is a classical valuation  $v_0$  such that  $v_0(B) = 1$  for every  $B \in \Sigma$  and  $v_0(A) = 0$ . Use the same assignment to atomic formulas and extend it homomorphically into **A**, obtaining a valuation  $v_{\mathcal{M}}$ . The operations of **A** agree with the classical truth functions on  $\{0, 1\}$ . An induction on formula complexity therefore gives  $v_{\mathcal{M}}(H) = v_0(H)$  for every formula  $H$ . Hence  $v_{\mathcal{M}}$  designates every member of  $\Sigma$  but not  $A$ , so  $\Sigma \not\models_{\mathcal{M}} A$ . ■

**Generalization.** The same argument works for every matrix  $M = \langle \mathbf{B}, D \rangle$  such that  $1 \in D, 0 \notin D$ , and the operations of **B** agree with the classical truth functions when restricted to  $\{0, 1\}$ . Every such matrix satisfies  $\Sigma \models_M A \implies \Sigma \models_{\text{CPL}} A$ .

## 5 CONSEQUENCE OPERATOR

**Definition. 1** We define a consequence operator  $Cn_M : \mathcal{P}(\text{For}) \longrightarrow \mathcal{P}(\text{For})$  as follows:  
 $A \in Cn_M(\Sigma)$  iff  $\Sigma \models_M A$ .

**Lemma.** The consequence operator has the following properties:

1.  $\Sigma \subseteq Cn_M(\Sigma)$ . (Extensivity)
2. If  $\Sigma \subseteq \Delta$ , then  $Cn_M(\Sigma) \subseteq Cn_M(\Delta)$ . (Monotonicity)
3.  $Cn_M(\Sigma) = Cn_M(Cn_M(\Sigma))$ . (Transitivity)
4.  $e[Cn_M(\Sigma)] \subseteq Cn_M(e[\Sigma])$  for all endomorphisms  $e$ . (Structurality)

**Definition 2.** For each structural consequence operator  $Cn_M$  and each  $X \subseteq \text{For}$ , we define a matrix

$$L_X = \langle \text{For}, Cn_M(X) \rangle,$$

called a **Lindenbaum matrix**. Furthermore, we can define matrix consequence relation as following:

$$\Sigma \models_{L_X} A \text{ iff for all endomorphism } e : \text{For} \longrightarrow \text{For} \text{ if } e[\Sigma] \subseteq Cn_M(X), \text{ then } e(A) \in Cn_M(X)$$

**Definition 3.** For any given consequence operator  $Cn_M$ , we define the **Lindenbaum bundle**

$$\text{LBM} = \{ \langle \text{For}, Cn_M(X) \rangle \mid X \subseteq \text{For} \},$$

the class of all Lindenbaum matrices. The consequence relation with respect to a Lindenbaum bundle is defined by

$$\Gamma \models_{\text{LBM}} A \text{ iff, for every } e \in \text{Hom}(\text{For}, \text{For}) \text{ and every } \langle \text{For}, Cn_M(X) \rangle \in \text{LBM}, \\ \text{if } e(B) \in Cn_M(X) \text{ for every } B \in \Gamma, \text{ then } e(A) \in Cn_M(X)$$

**Theorem 1. (Wójcicki)** Let  $\mathcal{L} = \langle \text{For}, Cn_M \rangle$  be a logic. Then, for every  $\Gamma \cup \{A\} \subseteq \text{For}$ ,

$$\Gamma \models_{\text{LBM}} A \text{ iff } A \in Cn_M(\Gamma).$$

**Theorem 2.**

$$Cn_M(\Sigma) = \bigcap \{ Cn_{L_X}(\Sigma) \mid L_X \in \text{LBM} \}.$$

## 6 SUSZKO'S THESIS

*Example.* Let  $M = \langle \{a, b, c, d\}, \text{et}, \text{vel}, \text{seq}^a, \text{non}\rangle, \{a, b\}$ . The designated values are  $a$  and  $b$ . We have following truth table:

| et       | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | vel      | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | seq      | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>non</i> |          |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|------------|----------|
| <i>a</i> | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>a</i>   | <i>d</i> |
| <i>b</i> | <i>a</i> | <i>a</i> | <i>c</i> | <i>d</i> | <i>b</i> | <i>a</i> | <i>b</i> | <i>b</i> | <i>b</i> | <i>b</i> | <i>a</i> | <i>b</i> | <i>c</i> | <i>c</i> | <i>b</i>   | <i>c</i> |
| <i>c</i> | <i>c</i> | <i>c</i> | <i>c</i> | <i>d</i> | <i>c</i> | <i>a</i> | <i>b</i> | <i>c</i> | <i>c</i> | <i>c</i> | <i>a</i> | <i>b</i> | <i>b</i> | <i>b</i> | <i>c</i>   | <i>b</i> |
| <i>d</i> | <i>d</i> | <i>d</i> | <i>d</i> | <i>d</i> | <i>d</i> | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>d</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>a</i> | <i>d</i>   | <i>a</i> |

We can show:  $\Sigma \models_M A$  iff  $\Sigma \models_{\text{CPL}} A$ .

\*CPL is classical propositional logic,  $\text{CPL} = \langle \{1, 0\}, \wedge, \vee, \rightarrow, \neg, \{1\} \rangle$ .

“Latin, sequitur/sequi, means “follow”

*Proof.* ( $\Rightarrow$ ) Assume that  $\Sigma \models_M A$ . Let  $v_c \in \text{Hom}(\mathcal{S}, \text{CPL})$  be an arbitrary valuation such that  $v_c(B) = 1$  for all  $B \in \Sigma$ . We need to show that  $v_c(A) = 1$ . Define  $g : \{0, 1\} \rightarrow \{a, b, c, d\}$  by  $g(1) = a$  and  $g(0) = d$ . Inspection of the truth tables shows that  $\{a, d\}$  is closed under all the operations of  $M$  and that  $g$  preserves  $\wedge, \vee, \rightarrow$ , and  $\neg$ . Hence  $g$  is a homomorphism.

Let  $v_m = g \circ v_c$ . Then  $v_m$  is a valuation into  $M$ . Since  $v_c(B) = 1$  for every  $B \in \Sigma$ , we have  $v_m(B) = a \in \{a, b\}$  for every  $B \in \Sigma$ . Since  $\Sigma \models_M A$ , it follows that  $v_m(A) \in \{a, b\}$ . The range of  $v_m$  is contained in  $\{a, d\}$ , so  $v_m(A) = a$ . Therefore  $v_c(A) = 1$ . Since  $v_c$  was arbitrary,  $\Sigma \models_{\text{CPL}} A$ .

( $\Leftarrow$ ) Assume that  $\Sigma \models_{\text{CPL}} A$ . Define  $h : \{a, b, c, d\} \rightarrow \{0, 1\}$  by  $h(a) = h(b) = 1$  and  $h(c) = h(d) = 0$ . Inspection of the truth tables shows that  $h$  preserves all the operations:  $h(\text{et}(x, y)) = h(x) \wedge h(y)$ ,  $h(\text{vel}(x, y)) = h(x) \vee h(y)$ ,  $h(\text{seq}(x, y)) = h(x) \rightarrow h(y)$ , and  $h(\text{non}(x)) = \neg h(x)$ . Hence  $h$  is a homomorphism.

Let  $v_m$  be an arbitrary valuation into  $M$  such that  $v_m(B) \in \{a, b\}$  for all  $B \in \Sigma$ . Then  $v_c = h \circ v_m$  is a classical valuation and  $v_c(B) = 1$  for all  $B \in \Sigma$ . Since  $\Sigma \models_{\text{CPL}} A$ , we have  $v_c(A) = 1$ . Thus  $h(v_m(A)) = 1$ , which means that  $v_m(A) \in \{a, b\}$ . Since  $v_m$  was arbitrary,  $\Sigma \models_M A$ . ■

**6.1 Suszko Reduction** Roman Suszko (1919–1979), at the 22nd Conference on the History of Logic in 1976, criticized Łukasiewicz’s many-valued project: “How was it possible that the humbug of many logical values persisted over the last fifty years?” It seems that, even if we consider the algebraic values of many-valued logic, we can still divide them into two classes: designated and non-designated values. Then it seems that it still two values: truth designated and falsity-non-designated. This indicates that we can reduce many-valued logic to two-valued logic.

**Definition. 1** Let  $M = \langle \mathcal{A}, D \rangle$  be a matrix, and let  $H$  be the class of all valuations into  $M$ . We define the class of all two-valued valuations  $\text{Tv}$  for  $M$  as follows:  $\text{Tv} = \{t_h \mid h \in H\}$ , where, for every formula  $A$ ,  $t_h(A) = 1$  if  $h(A) \in D$ , and  $t_h(A) = 0$  if  $h(A) \notin D$ :

$$t_h(A) = \begin{cases} 1, & \text{if } h(A) \in D, \\ 0, & \text{if } h(A) \notin D. \end{cases}$$

We have  $\Sigma \models_{\text{Tv}} A$  iff, for every  $t_h \in \text{Tv}$ , if  $t_h(B) = 1$  for all  $B \in \Sigma$ , then  $t_h(A) = 1$ .

**Theorem.**  $\Sigma \models_M A$  iff  $\Sigma \models_{Tv} A$ .

*Proof.* By definition,  $\Sigma \models_M A$  iff, for every valuation  $h \in H$ , if  $h(B) \in D$  for all  $B \in \Sigma$ , then  $h(A) \in D$ . For every  $h \in H$  and every formula  $B$ , we have  $h(B) \in D$  iff  $t_h(B) = 1$ . Therefore, the preceding condition holds iff, for every  $t_h \in Tv$ , if  $t_h(B) = 1$  for all  $B \in \Sigma$ , then  $t_h(A) = 1$ . By definition, this is equivalent to  $\Sigma \models_{Tv} A$ . Hence  $\Sigma \models_M A$  iff  $\Sigma \models_{Tv} A$ . ■

**Corollary.** Every Tarskian consequence relation is inferentially two-valued.

*Example.* Consider  $L3$ .

| $x$           | $\neg x$      | $\rightarrow$ | 1 | $\frac{1}{2}$ | 0             |
|---------------|---------------|---------------|---|---------------|---------------|
| 1             | 0             | 1             | 1 | $\frac{1}{2}$ | 0             |
| $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | 1             | $\frac{1}{2}$ |
| 0             | 1             | 0             | 1 | 1             | 1             |

Passing into  $Tv$  sends 1 to 1, and both  $\frac{1}{2}$  and 0 to 0. The tables therefore have the following two-valued form:

| $v(A)$ | $v(\neg A)$ | $\rightarrow$ | 1 | 0 | 0 |
|--------|-------------|---------------|---|---|---|
| 1      | 0           | 1             | 1 | 0 | 0 |
| 0      | 0           | 0             | 1 | 1 | 0 |
| 0      | 1           | 0             | 1 | 1 | 1 |

Compared with classical logic, these valuations satisfy:

1.  $v(A) = 0$  or  $v(\neg A) = 0$ .
2. If  $v(B) = 1$ , then  $v(A \rightarrow B) = 1$ .
3. If  $v(A) = 1$  and  $v(B) = 0$ , then  $v(A \rightarrow B) = 0$ .
4. If  $v(A) = v(B)$  and  $v(\neg A) = v(\neg B)$ , then  $v(A \rightarrow B) = 1$ .
5. If  $v(A) = v(B) = 0$  and  $v(\neg A) \neq v(\neg B)$ , then  $v(A \rightarrow B) = v(\neg A)$ .
6. If  $v(\neg A) = 0$ , then  $v(\neg\neg A) = v(A)$ .
7. If  $v(A) = 1$  and  $v(B) = 0$ , then  $v(\neg(A \rightarrow B)) = v(\neg B)$ .
8. If  $v(A) = v(\neg A) = v(B)$  and  $v(\neg B) = 1$ , then  $v(\neg(A \rightarrow B)) = 0$ .

Equivalently, the three algebraic values are represented by pairs of inferential values:

$$\left\{1, \frac{1}{2}, 0\right\} \longrightarrow \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}.$$

**Definition.** Let  $\mathcal{L} = \langle \text{For}, \text{Cn} \rangle$  be a logic. We call a function  $v : \text{For} \rightarrow \{0, 1\}$  a **bivaluation**. Given a set  $V$  of bivaluations, we define  $\Sigma \models_V A$  iff, for every  $v \in V$ , if  $v(B) = 1$  for all  $B \in \Sigma$ , then  $v(A) = 1$ . The logic  $\mathcal{L}$  is called **bivalent** iff there exists a set  $V$  of bivaluations such that, for all  $A$  and  $\Sigma$ ,  $A \in \text{Cn}(\Sigma)$  iff  $\Sigma \models_V A$ .

**Lemma.** Let  $\mathcal{L} = \langle \text{For}, \text{Cn} \rangle$  be a logic. Then  $\text{Cn}$  is reflexive iff  $\mathcal{L}$  is bivalent.

*Proof.* ( $\Rightarrow$ ) Suppose that  $\text{Cn}$  is reflexive. For every  $X \subseteq \text{For}$ , let  $\varphi_X$  be the characteristic function of  $\text{Cn}(X)$ , and define

$$V_\varphi = \{\varphi_X \mid X \subseteq \text{For}\}, \quad \varphi_X(B) = \begin{cases} 1, & \text{if } B \in \text{Cn}(X), \\ 0, & \text{if } B \notin \text{Cn}(X). \end{cases}$$

We show that, for every  $\Sigma \subseteq \text{For}$  and every  $A \in \text{For}$ ,  $\Sigma \models_{V_\varphi} A$  iff  $A \in \text{Cn}(\Sigma)$ .

First, suppose that  $A \notin \text{Cn}(\Sigma)$ . By the definition of  $\varphi_\Sigma$ , we have  $\varphi_\Sigma(A) = 0$ . Since  $\text{Cn}$  is reflexive,  $\Sigma \subseteq \text{Cn}(\Sigma)$ , and hence  $\varphi_\Sigma(B) = 1$  for every  $B \in \Sigma$ . Thus  $\varphi_\Sigma$  is a countervaluation to  $\Sigma \models_{V_\varphi} A$ , so  $\Sigma \not\models_{V_\varphi} A$ . Therefore

$$\Sigma \models_{V_\varphi} A \implies A \in \text{Cn}(\Sigma).$$

Conversely, suppose that  $A \in \text{Cn}(\Sigma)$ . Let  $\varphi_X \in V_\varphi$  be arbitrary and suppose that  $\varphi_X(B) = 1$  for every  $B \in \Sigma$ . By the definition of  $\varphi_X$ , this means that  $\Sigma \subseteq \text{Cn}(X)$ . By the previously established monotonicity and transitivity of  $\text{Cn}$ ,

$$\text{Cn}(\Sigma) \subseteq \text{Cn}(\text{Cn}(X)) = \text{Cn}(X).$$

Since  $A \in \text{Cn}(\Sigma)$ , it follows that  $A \in \text{Cn}(X)$ , and therefore  $\varphi_X(A) = 1$ . Since  $\varphi_X$  was arbitrary, we obtain  $\Sigma \models_{V_\varphi} A$ . Hence

$$A \in \text{Cn}(\Sigma) \implies \Sigma \models_{V_\varphi} A.$$

Consequently,  $\Sigma \models_{V_\varphi} A$  iff  $A \in \text{Cn}(\Sigma)$ , so  $\mathcal{L}$  is bivalent.

( $\Leftarrow$ ) Suppose that  $\mathcal{L}$  is bivalent. Then there is a set  $V$  of bivaluations such that, for every  $\Sigma \subseteq \text{For}$  and every  $A \in \text{For}$ ,

$$\Sigma \models_V A \text{ iff } A \in \text{Cn}(\Sigma).$$

Let  $A \in \Sigma$ . Every bivaluation that assigns 1 to every formula in  $\Sigma$  assigns 1 to  $A$ , since  $A$  itself belongs to  $\Sigma$ . Therefore  $\Sigma \models_V A$ . By bivalence,  $A \in \text{Cn}(\Sigma)$ . Hence  $\Sigma \subseteq \text{Cn}(\Sigma)$ , and so  $\text{Cn}$  is reflexive. ■

This proof was provided by Marcelo Tsuji (1998)

## References for Chapter 6

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## 7 Q-LOGIC AND P-LOGIC

**7.1 q-logic** Q-logic was introduced by Malinowski (1993). Q here means "quasi". We can regard the elements of  $D$  as accepted values and the elements of  $R$  as rejected values. Q-consequence leads from non-rejected assumptions to accepted conclusions.

**Definition. 1** A q-matrix  $M = \langle A, D, R \rangle$  is a tuple consisting of an algebra  $A$  together with two sets  $D \subseteq A$  and  $R \subseteq A$  such that:

1.  $D \cup R \subseteq A$ ;
2.  $D \cap R = \emptyset$ ;
3.  $D \neq \emptyset$  and  $R \neq \emptyset$ .

**Definition 2.** A q-logic  $L_q = \langle \text{For}, \models_q \rangle$ , where  $\text{For}$  is a sentential algebra and  $\models_q$  is a q-consequence relation defined as follows: For  $\Sigma \cup \{A\} \subseteq \text{For}$ ,

$$\Sigma \models_q A \quad \text{iff} \quad \text{for every } v, v[\Sigma] \subseteq \text{ implies } v(A) \in D.$$

*Example.* Consider the q-matrix  $\mathbf{qL}_3 = \langle \{1, \frac{1}{2}, 0\}, \{1 - x, \min\{x, y\}, \max\{x, y\}\}, \{1\}, \{0\} \rangle$ , where  $D = \{1\}$  is the set of accepted values and  $R = \{0\}$  is the set of rejected values. We have  $p, p \rightarrow q \not\models_q q$ , where  $p \rightarrow q$  is defined as  $\neg p \vee q$ , since, for a valuation  $v$  such that  $v(p) = \frac{1}{2}$  and  $v(q) = 0$ , we have  $v(p \rightarrow q) = \frac{1}{2}$ , and hence  $v[\{p, p \rightarrow q\}] \cap R = \emptyset$ , while  $v(q) = 0 \notin D$ . Moreover,  $\not\models_q p \vee \neg p$ , since, if  $v(p) = \frac{1}{2}$ , then  $v(p \vee \neg p) = \frac{1}{2} \notin D$ .

q-logics are not generally reflexive, that is,  $A \not\models_q A$  for some formula  $A$  in some q-logic.

*Proof.* Assume that  $p \models_q p$ . Then, for every valuation  $v$ , if  $v(p) \notin R$ , then  $v(p) \in D$ . Since  $D \cup R \subseteq A$ , there exists some  $a \in A$  such that  $a \notin D$  and  $a \notin R$ . Choose a valuation  $v$  such that  $v(p) = a$ . Then  $v(p) \notin R$ , so our assumption implies that  $v(p) \in D$ . However, by the choice of  $a$ , we have  $v(p) \notin D$ , which is a contradiction. Therefore,  $p \not\models_q p$ , and hence q-logics are not generally reflexive. ■

**q-Cut.** If  $\Sigma \models_q A$  and  $\Delta, A \models_q B$ , then

$$\Sigma, \Delta \models_q B.$$

*Proof.* Suppose that  $\Sigma \models_q A$  and  $\Delta, A \models_q B$ , but  $\Sigma, \Delta \not\models_q B$ . Then there is a valuation  $v$  such that

$$v[\Sigma \cup \Delta] \cap R = \emptyset \quad \text{and} \quad v(B) \notin D.$$

Since  $v[\Sigma] \cap R = \emptyset$  and  $\Sigma \models_q A$ , we have  $v(A) \in D$ . Because  $D \cap R = \emptyset$ ,  $v(A) \notin R$ . Hence

$$v[\Delta \cup \{A\}] \cap R = \emptyset.$$

Since  $\Delta, A \models_q B$ , it follows that  $v(B) \in D$ , a contradiction. Therefore  $\Sigma, \Delta \models_q B$ . ■

**7.2 p-logic** P-logic was introduced by Frankowski (2004). P means "plausible". In that sense, p-consequence means to a non-rejected conclusion follows from designated premises.



**Definition 3.** A p-logic  $L_p = \langle \text{For}, \models_p \rangle$  consists of a sentential algebra For and a p-consequence relation  $\models_p$ , defined as follows:

$$\Gamma \models_p A \quad \text{iff} \quad \text{for every valuation } v, v[\Gamma] \subseteq D \text{ implies } v(A) \notin R.$$



p-logic is reflexive



p-logics are not transitive



*Counter-example for p-Cut.* Let

$$\mathcal{M}_o = \langle \langle \{1, \frac{1}{2}, 0\}, o \rangle, \{1\}, \{0\} \rangle,$$

where

| $A$           | $oA$          |
|---------------|---------------|
| 1             | $\frac{1}{2}$ |
| $\frac{1}{2}$ | $\frac{1}{2}$ |
| 0             | $\frac{1}{2}$ |

Then

$$A \models_p oA \quad \text{and} \quad oA \models_p B,$$

but

$$A \not\models_p B.$$

Indeed,  $oA$  is never rejected and is never accepted. The last inference fails when  $v(A) = 1$  and  $v(B) = 0$ . Thus

$$\Sigma \models_p A \quad \text{and} \quad \Delta, A \models_p B \quad \not\Rightarrow \quad \Sigma, \Delta \models_p B.$$



**Theorem.** For the matrix  $\mathcal{M} = \langle \langle \{1, \frac{1}{2}, 0\}, 1 - x, \max(x, y), \min(x, y) \rangle, \{1\}, \{0\} \rangle$ .

$$\Sigma \models_p H \quad \text{iff} \quad \Sigma \models_{\text{CL}} H.$$

*Proof.* ( $\Rightarrow$ ) Suppose that  $\Sigma \not\models_{\text{CL}} H$ . Then there is a classical valuation  $v_{\text{CL}}$  such that  $v_{\text{CL}}(A) = 1$  for every  $A \in \Sigma$  and  $v_{\text{CL}}(H) = 0$ . This is also a valuation in  $\mathcal{M}$ , with

$$v_{\text{CL}}[\Sigma] \subseteq D \quad \text{and} \quad v_{\text{CL}}(H) \in R.$$

Hence  $\Sigma \not\models_p H$ .

( $\Leftarrow$ ) Suppose that  $\Sigma \not\models_p H$ . Then there is a valuation  $v_p$  such that

$$v_p(A) = 1 \text{ for every } A \in \Sigma, \quad v_p(H) = 0.$$

Let  $t : \{1, \frac{1}{2}, 0\} \rightarrow \{0, 1\}$  be the mapping

$$t(1) = 1, \quad t\left(\frac{1}{2}\right) = 0, \quad t(0) = 0.$$

Use  $t \circ v_p$  on the propositional variables and extend it as a classical valuation. Boolean sharpening preserves every formula that has value 1 or 0 under the Strong Kleene operations. Therefore the resulting classical valuation assigns 1 to all formulas in  $\Sigma$  and 0 to  $H$ . Hence  $\Sigma \not\equiv_{\text{CL}} H$ . ■

### 7.3 $k$ -valued logic

**Definition 4.** A logical system is *logically* (or *inferentially*)  $k$ -valued if it is a tuple

$$\mathcal{L} = \langle \overline{\text{For}}, \models_1, \dots, \models_k \rangle,$$

where  $\models_1, \dots, \models_k$  are pairwise distinct, canonically defined consequence relations on the language.

**Definition 5.** Let  $\mathbf{M} = \langle \mathbf{A}, D \rangle$  be a logical matrix, where  $D$  is the set of designated values. The ordinary matrix consequence relation can be stated as forward preservation of designated values:

$$\Gamma \models_D A \iff \text{for every valuation } v, \text{ if } v[\Gamma] \subseteq D, \text{ then } v(A) \in D.$$

The same relation can be stated backward as preservation of undesigned values:

$$\Gamma \models_D A \iff \text{for every valuation } v, \text{ if } v(A) \notin D, \text{ then } v(B) \notin D \text{ for some } B \in \Gamma.$$

*Fact.* Forward preservation and backward preservation are contrapositively equivalent. Hence classical logic, LP,  $K_3$ , and FDE can all be presented in both ways: if a consequence holds by forward preservation, it also holds by backward preservation.

**Does classical logic really have two values?** Ordinary classical consequence preserves the designated value 1 forward. Equivalently, it preserves the undesigned value 0 backward. These are two formulations of one consequence relation, rather than two independent relations. A distinct relation that preserves falsity forward can be defined by

$$\Gamma \models_F A \iff \text{for every classical valuation } v, \text{ if } v[\Gamma] \subseteq \{0\}, \text{ then } v(A) = 0.$$

### References for Chapter 7

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## 8 FDE AND SIXTEEN3

**8.1 FDE** First-Degree Entailment logic was proposed by Nuel D. Belnap and J. Michael Dunn (1977). It is also called Belnap–Dunn Logic. The preliminary motivation for this logic is related to relevance logic E (1962). First-degree entailment means that, in  $A \rightarrow B$ ,  $A$  and  $B$  contain only negation, conjunction, and disjunction. We now understand FDE as a four-valued logic that combines LP and K3: LP can be interpreted as allowing propositions to be both true and false, while K3 can be interpreted as allowing propositions to be neither true nor false.

**Definition.** The matrix for **First-Degree Entailment (FDE)** is

$$\mathcal{M}_{\text{FDE}} = \langle \langle \{t, b, n, f\}, \widetilde{\wedge}, \widetilde{\vee}, \widetilde{\neg} \rangle, \{t, b\} \rangle,$$

where  $t = \{1\}$ ,  $b = \{1, 0\}$ ,  $n = \emptyset$ , and  $f = \{0\}$ . The designated values are  $t$  and  $b$ . The truth tables are as follows:

| $\widetilde{\wedge}$ | $t$ | $b$ | $n$ | $f$ | $\widetilde{\vee}$ | $t$ | $b$ | $n$ | $f$ | $A$ | $\widetilde{\neg}A$ |
|----------------------|-----|-----|-----|-----|--------------------|-----|-----|-----|-----|-----|---------------------|
| $t$                  | $t$ | $b$ | $n$ | $f$ | $t$                | $t$ | $t$ | $t$ | $t$ | $t$ | $f$                 |
| $b$                  | $b$ | $b$ | $f$ | $f$ | $b$                | $t$ | $b$ | $t$ | $b$ | $b$ | $b$                 |
| $n$                  | $n$ | $f$ | $n$ | $f$ | $n$                | $t$ | $t$ | $n$ | $n$ | $n$ | $n$                 |
| $f$                  | $f$ | $f$ | $f$ | $f$ | $f$                | $t$ | $b$ | $n$ | $f$ | $f$ | $t$                 |

In FDE, the four values are represented by the subsets of  $\{1, 0\}$ :

$$\mathbf{t} = \{1\}, \quad \mathbf{b} = \{1, 0\}, \quad \mathbf{n} = \emptyset, \quad \mathbf{f} = \{0\}.$$

They may be read as true only, both true and false, neither true nor false, and false only, respectively. We can provide following semantics:

**Definition 2.** Let  $v : \text{FORM} \rightarrow \mathcal{P}(\{1, 0\})$  be a function from the set of all formulas to the set of truth values of BD. The function  $v$  is called a **valuation** iff, for all  $A, B \in \text{FORM}$ , the following conditions are satisfied:

$$\begin{aligned} 1 \in v(\neg A) &\iff 0 \in v(A), \\ 0 \in v(\neg A) &\iff 1 \in v(A), \\ 1 \in v(A \wedge B) &\iff 1 \in v(A) \text{ and } 1 \in v(B), \\ 0 \in v(A \wedge B) &\iff 0 \in v(A) \text{ or } 0 \in v(B), \\ 1 \in v(A \vee B) &\iff 1 \in v(A) \text{ or } 1 \in v(B), \\ 0 \in v(A \vee B) &\iff 0 \in v(A) \text{ and } 0 \in v(B). \end{aligned}$$

Actually, we can use parital order to describe the truth values.

**Definition 3.** Let  $v : \text{FORM} \rightarrow \mathcal{P}(\{1, 0\})$ . For every  $x \in \text{FORM}$ , define

$$\begin{aligned} x^t &:= \{z \in v(x) \mid z = 1\}, \\ x^f &:= \{z \in v(x) \mid z = 0\}. \end{aligned}$$

For all  $A, B \in \text{FORM}$ , the **truth order**  $\leq_t$  and the dual **falsity order**  $\leq_f$  are defined by

$$\begin{aligned} v(A) \leq_t v(B) &\iff A^t \subseteq B^t \text{ and } B^f \subseteq A^f, \\ v(A) \leq_f v(B) &\iff A^f \subseteq B^f \text{ and } B^t \subseteq A^t. \end{aligned}$$

| $x$                      | $x^t$       | $x^f$       |
|--------------------------|-------------|-------------|
| $\mathbf{t} = \{1\}$     | $\{1\}$     | $\emptyset$ |
| $\mathbf{b} = \{1, 0\}$  | $\{1\}$     | $\{0\}$     |
| $\mathbf{n} = \emptyset$ | $\emptyset$ | $\emptyset$ |
| $\mathbf{f} = \{0\}$     | $\emptyset$ | $\{0\}$     |

FDE is also a highly useful logic with important applications in computer science. In particular, it provides an information ordering that represents how much information a system has. Under this ordering, N contains the least information, B contains the most information, and T and F each contain only partial information. This ordering is especially useful for modelling how information from different sources can be accumulated, including cases in which the sources provide incomplete or conflicting information.

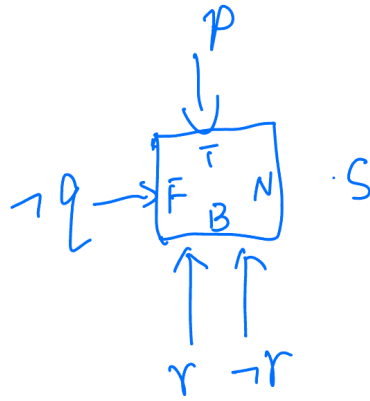


Figure 1: How a computer should think

**Definition 4.** For all  $A, B \in \text{FORM}$ , the **information order**  $\leq_i$  is defined by

$$v(A) \leq_i v(B) \iff v(A) \subseteq v(B).$$

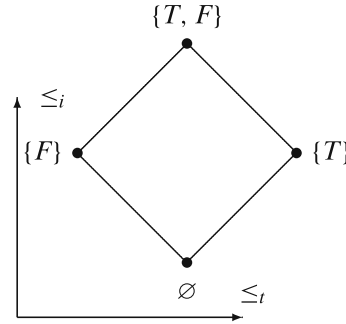


Figure 2: The bilattice  $\text{FOUR}_2$ . The vertical axis represents the information order  $\leq_i$ , and the horizontal axis represents the truth order  $\leq_t$  (Wansing 2010, Fig. 1).

**8.2 SIXTEEN3** The four FDE values are the members of the powerset  $\mathbf{4} := \mathcal{P}(\{0, 1\}) = \{\mathbf{n}, \mathbf{f}, \mathbf{t}, \mathbf{b}\}$ , or  $\{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$ . The construction with sixteen values applies the powerset operation once more:

$$\mathbf{16} := \mathcal{P}(\mathbf{4}) = \mathcal{P}(\mathcal{P}(\{0, 1\})).$$

Thus an element of  $\mathbf{16}$  is a set of FDE values. Using the notation of the lecture notes, the sixteen generalized truth values are as follows:

- |                                               |                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------|
| 1. $\emptyset = \{\}$                         | 9. $\mathbf{FT} = \{\mathbf{f}, \mathbf{t}\}$                                      |
| 2. $\mathbf{N} = \{\mathbf{n}\}$              | 10. $\mathbf{FB} = \{\mathbf{f}, \mathbf{b}\}$                                     |
| 3. $\mathbf{F} = \{\mathbf{f}\}$              | 11. $\mathbf{TB} = \{\mathbf{t}, \mathbf{b}\}$                                     |
| 4. $\mathbf{T} = \{\mathbf{t}\}$              | 12. $\mathbf{NFT} = \{\mathbf{n}, \mathbf{f}, \mathbf{t}\}$                        |
| 5. $\mathbf{B} = \{\mathbf{b}\}$              | 13. $\mathbf{NFB} = \{\mathbf{n}, \mathbf{f}, \mathbf{b}\}$                        |
| 6. $\mathbf{NF} = \{\mathbf{n}, \mathbf{f}\}$ | 14. $\mathbf{NTB} = \{\mathbf{n}, \mathbf{t}, \mathbf{b}\}$                        |
| 7. $\mathbf{NT} = \{\mathbf{n}, \mathbf{t}\}$ | 15. $\mathbf{FTB} = \{\mathbf{f}, \mathbf{t}, \mathbf{b}\}$                        |
| 8. $\mathbf{NB} = \{\mathbf{n}, \mathbf{b}\}$ | 16. $\mathbf{A} = \{\mathbf{n}, \mathbf{f}, \mathbf{t}, \mathbf{b}\} = \mathbf{4}$ |

| Cardinality | Generalized truth values                                                                                                                                                 |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0           | $\emptyset$                                                                                                                                                              |
| 1           | $\{\mathbf{n}\}, \{\mathbf{t}\}, \{\mathbf{f}\}, \{\mathbf{b}\}$                                                                                                         |
| 2           | $\{\mathbf{n}, \mathbf{t}\}, \{\mathbf{n}, \mathbf{f}\}, \{\mathbf{n}, \mathbf{b}\}, \{\mathbf{t}, \mathbf{f}\}, \{\mathbf{t}, \mathbf{b}\}, \{\mathbf{f}, \mathbf{b}\}$ |
| 3           | $\{\mathbf{n}, \mathbf{t}, \mathbf{f}\}, \{\mathbf{n}, \mathbf{t}, \mathbf{b}\}, \{\mathbf{n}, \mathbf{f}, \mathbf{b}\}, \{\mathbf{t}, \mathbf{f}, \mathbf{b}\}$         |
| 4           | $\{\mathbf{n}, \mathbf{t}, \mathbf{f}, \mathbf{b}\}$                                                                                                                     |

This 16-valued logic was introduced by Yaroslav Shramko and Heinrich Wansing (2005). The motivation is to follow FDE in modeling information processing, like how a computer network processes information (four computers). Similarly, we can use the three orders defined above to describe SIXTEEN3.

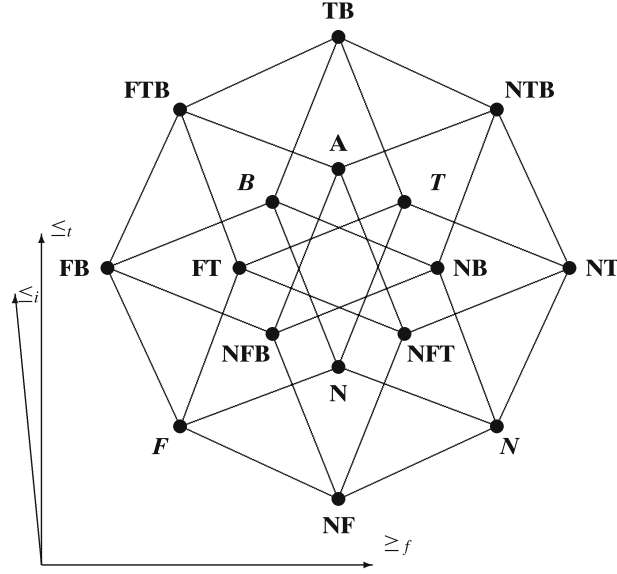


Figure 3: The trilattice  $\text{SIXTEEN}_3$ . The information ordering is read as coming out of the page (Wansing 2010, Fig. 2).

**Definition 5.** Let  $v : \text{FORM} \longrightarrow \mathcal{PP}(\{1, 0\})$ . For every  $x \in \text{FORM}$ , define

$$\begin{aligned} x^t &:= \{z \in v(x) \mid 1 \in z\}, \\ x^{-t} &:= \{z \in v(x) \mid 1 \notin z\}, \\ x^f &:= \{z \in v(x) \mid 0 \in z\}, \\ x^{-f} &:= \{z \in v(x) \mid 0 \notin z\}. \end{aligned}$$

For all  $A, B \in \text{FORM}$ , the **truth order**  $\leq_t$  and the dual **falsity order**  $\leq_f$  are defined by

$$\begin{aligned} v(A) \leq_t v(B) &\iff A^t \subseteq B^t \text{ and } B^{-t} \subseteq A^{-t}, \\ v(A) \leq_f v(B) &\iff A^f \subseteq B^f \text{ and } B^{-f} \subseteq A^{-f}. \end{aligned}$$

The three orders induce three consequence relations:

$$\begin{aligned} A \vDash_t B &\iff v(A) \leq_t v(B) \text{ for every valuation } v, \\ A \vDash_f B &\iff v(A) \leq_f v(B) \text{ for every valuation } v, \\ A \vDash_i B &\iff v(A) \leq_i v(B) \text{ for every valuation } v. \end{aligned}$$

Accordingly, the logic with sixteen values can be represented by a matrix with three canonical consequence relations

$$\mathcal{L}_{16} = \langle \overline{\text{For}}, \vDash_t, \vDash_f, \vDash_i \rangle.$$

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